

Naga Sai Sindhura Pandrangi

Data Engineer | Data Analyst | Data Scientist / ML Engineer

Fremont, CA | (669) 260-0662 | sindhura.naga98@gmail.com | linkedin.com/in/sindhurapandrangi | nagasaisindhura.github.io

Professional Summary

Data professional with 5+ years of experience across data engineering, analytics, and machine learning. Builds production ETL/ELT pipelines and dimensional models on **Snowflake**, **GCP (BigQuery, Cloud Storage)**, and SQL Server using **dbt**, **Apache Airflow**, Python, and SQL, and delivers KPI reporting in **Power BI**, **Tableau**, and **Looker Studio**. Hands-on with the ML lifecycle: feature engineering, model training and evaluation, **LLM fine-tuning**, **RAG pipelines**, and **multi-agent systems**, with models deployed through FastAPI, Flask, and Docker.

Technical Skills

Data Warehousing: Snowflake, Google BigQuery, MS SQL Server, MySQL

ETL/ELT & Transformation: dbt (Core & Cloud), Jinja Macros, Incremental Models, dbt Tests, ETL/ELT Pipeline Design, Star Schema & Dimensional Modeling

Orchestration: Apache Airflow (DAG design, scheduling, monitoring, alerting), Cloud Composer

Cloud (GCP): BigQuery, Google Cloud Storage, Cloud Composer, REST API Ingestion

Visualization & BI: Power BI (DAX, Power Query), Tableau, Looker Studio

Languages: Python (Pandas, NumPy, Scikit-learn), SQL (CTEs, window functions, query optimization), Bash

Analytics: KPI Design, Exploratory Data Analysis, Data Cleaning, Data Validation, Plotly, Seaborn, Matplotlib

Machine Learning: Random Forest, XGBoost, Regression, Feature Engineering, PyTorch, CNN/BiLSTM, Model Evaluation (Accuracy, ROC-AUC, BLEU, ROUGE, BERTScore)

GenAI: RAG Pipelines, LLM Fine-Tuning, Hugging Face Transformers, Multi-Agent Systems, Vector Search, Prompt Engineering, Quantization

DevOps & Tools: Git, GitHub/GitLab, CI/CD, Docker, FastAPI, Flask, Streamlit, Jira, Confluence

Professional Experience

AI/ML Engineer

Feb 2026 – Present

Referu.ai

Remote

- Built and fine-tuned domain-specific LLMs on curated intake data, improving response relevance and grounding for a multi-agent AI platform serving users across multiple jurisdictions.
- Designed and deployed end-to-end **RAG pipelines** covering document ingestion, chunking, embedding generation, vector indexing, and retrieval re-ranking, so agents answer from verified, jurisdiction-specific sources.
- Developed **multi-agent workflows** in which specialized agents handle intake, retrieval, reasoning, and compliance checks, using tool calling and structured outputs.
- Trained and evaluated models using supervised fine-tuning and prompt optimization, tracking groundedness, accuracy, and hallucination rates to guide each iteration.
- Added automated evaluation, testing, and monitoring to the model pipeline, cutting non-compliant model outputs by **27%**.
- Analyzed retrieval and intake data to identify failure patterns, then fixed chunking and retrieval logic, improving grounded-response accuracy in production.

Data Engineer

Sep 2024 – Jul 2025

MMP Insulators

Remote

- Built **Apache Airflow** DAGs to ingest sales, inventory, and CRM data from 5+ regional systems into **BigQuery on GCP**, cutting manual data collection effort by **35%**.
- Modeled a star schema in BigQuery using **dbt** (staging, intermediate, and mart layers) and optimized SQL transformations, reducing dashboard query time by **30%** and reporting errors by **20%**.
- Wrote dbt tests and documentation for key models, catching data issues before they reached reports and improving trust in the numbers.
- Built and maintained **15+ Tableau dashboards** on the warehouse, giving leadership real-time visibility into regional and product-level sales trends.
- Partnered with sales and operations stakeholders to define KPIs and reporting requirements, and managed code through Git with peer review in an Agile team using Jira.

Data Engineer

S&P Global

Aug 2022 – Aug 2023

Remote

- Built Python and SQL ETL pipelines loading structured and unstructured financial data from multiple source systems into **Snowflake**, raising data integrity to **99.5%** and improving data quality by **25%**.
- Modeled dimensional schemas (facts and dimensions) in Snowflake using **dbt**, with reusable Jinja macros, incremental models, and automated tests for consistent, auditable transformations.
- Orchestrated daily and intraday loads with **Apache Airflow**, adding dependency management, retries, and failure alerts to keep financial datasets fresh and reliable.
- Tuned Snowflake workloads through warehouse sizing, clustering, and query rewrites, improving query performance for downstream analysts.
- Built **10+ Power BI dashboards** with DAX measures for financial KPIs, cutting stakeholder reporting turnaround by **20%**.
- Documented data lineage, mappings, and business rules to support audits and onboarding of new team members.

Data Engineer – DaVita Kidney Care Project

Tata Consultancy Services (TCS)

Jun 2019 – Jul 2022

India

- Built and maintained **Apache Airflow**-orchestrated ETL pipelines on **GCP** that ingested patient and treatment data from multiple clinical source systems into a **BigQuery** warehouse.
- Designed fact and dimension tables in BigQuery with partitioning and clustering, supporting clinical and operational reporting at scale.
- Added Airflow monitoring, SLAs, and alerting that cut incident response time by **50%**, and refactored data jobs to reduce pipeline runtime by **40%**.
- Wrote complex SQL (CTEs, window functions, aggregations) for transformation, reconciliation, and data validation, ensuring accuracy of healthcare data handled under strict compliance requirements.
- Supported downstream reporting in **Power BI**, working with business analysts to translate requirements into reliable, query-ready data models.
- Followed Git-based version control, code reviews, and release processes in an Agile delivery model.

Projects

AI Mock Interviewer for Data Roles (LLMs & RAG) *Hugging Face, RAG, FastAPI, Docker, GCP, Looker Studio*

- Fine-tuned four LLMs on **4,500+ Q&A pairs** and built a **RAG pipeline** that generates personalized interview questions from resumes and job descriptions, with real-time feedback.
- Evaluated models with BLEU, ROUGE, BERTScore, accuracy, and hallucination metrics; applied quantization for approximately **40% lower memory use** and low-latency inference.
- Built **Looker Studio** performance dashboards on GCP data pipelines, and shipped a full-stack app with a React frontend (Vercel) and FastAPI backend (Docker).

Automated ETL & KPI Dashboard on GCP

Airflow, dbt, BigQuery, GCS, Power BI

- Designed an ETL pipeline processing **100K+ daily records** from the New York Times REST APIs, landing raw data in Google Cloud Storage and transforming it with **dbt** into analytics-ready models.
- Built a star schema with **10+ fact and dimension tables** in BigQuery for faster querying and optimized reporting.
- Automated runs with **Airflow DAGs**, achieving **95%+ on-time data delivery**, and published auto-refreshing **Power BI** KPI dashboards connected directly to BigQuery.

Cricket Match Win Prediction

Python, Scikit-learn, XGBoost, Flask, Plotly

- Engineered features (runs left, balls left, wickets left) from **2,389 ODI matches** (2001–2024) and trained Random Forest, XGBoost, and Linear Regression models, reaching up to **91.6% accuracy and 0.97 AUC**.
- Deployed the model in a Flask web app for real-time win prediction, and produced 10+ EDA visualizations (Plotly, Seaborn, Matplotlib) to explain model behavior to stakeholders.

DeepFake Video Detection

PyTorch, CNN-BiLSTM, MTCNN, Deep Learning

- Developed a hybrid CNN-BiLSTM model (XceptionNet and ResNet-Swish-BiLSTM) achieving **97.01% validation accuracy and 0.988 ROC-AUC** on the Meta DFDC dataset.
- Built an MTCNN frame-extraction pipeline sampling 10 frames per video, and reduced overfitting with Focal Loss,

label smoothing, mixed-precision training, and cosine annealing with warm restarts.

Agentic Natural Language to SQL Translation

Hugging Face, Streamlit, Plotly, SQL

- Fine-tuned a Hugging Face LLM on the Spider dataset, improving BLEU from **0.38 to 0.52**, with an agent that asks follow-up questions to resolve vague queries before generating SQL.
- Built a Streamlit interface for schema exploration, query generation, and Plotly visualization for non-technical users.

California Road Safety Dashboard

Power BI, Power Query, DAX, MySQL

- Cleaned and modeled California accident data with Power Query and built interactive **Power BI** dashboards (slicers, drilldowns, time filters) on a MySQL source.
- Created DAX measures for KPIs such as accident frequency by county, average severity, and time-of-day trends, helping users identify high-risk zones.

Education

San Jose State University, San Jose, CA

Jan 2024 – Dec 2025

Master of Science in Applied Data Intelligence | GPA: 3.8/4.0

Jawaharlal Nehru Technological University Hyderabad (JNTUH), India

May 2015 – May 2019

Bachelor of Technology in Electronics & Communication Engineering | GPA: 3.6/4.0